Functional−group responses to field sunlight exposure

Sunlight exposure S+ UV− S−

Relative abundance

Alpha−glucan and fructan processing
q = 0.766

0.00200
0.00175
0.00150
0.00125

Aromatic−compound catabolism
q = 0.096

0.0045
0.0040
0.0035

Carotenoid biosynthesis and modification
q = 0.096

0.0024
0.0022
0.0020
0.0018
0.0016

Cellulose degradation
q = 0.165

1e−03
8e−04
6e−04
4e−04

Cutin degradation
q = 0.165

0.00015
0.00010
0.00005
0.00000

Enzymatic ROS defense
q = 0.165

0.0070
0.0065
0.0060

General DNA−damage repair and tolerance
q = 0.096

0.027
0.026
0.025
0.024

Hemicellulose degradation
q = 0.096

0.00225
0.00200
0.00175
0.00150

Lignin−associated oxidation
q = 0.412

8e−05
6e−05
4e−05
2e−05
0e+00

Oligosaccharide and disaccharide utilization
q = 0.147

0.0045
0.0040
0.0035
0.0030

Pectin degradation
q = 0.412

0.0015
0.0010
0.0005

Photoreactivation and nucleotide−excision repair
q = 0.870

0.00575
0.00550
0.00525
0.00500

Sensory photoreception
q = 0.165

1e−04
5e−05
0e+00

S+ UV− S−